Connector Cheat Sheet

Nov 12, 2026 · Bardia Mojra · CSE Senior Design Lab, UTA

Pick a connector by the continuous current it must carry, then match the wire gauge to it. Part of the lab's battery cheat sheets, with Battery Fundamentals, Charger Guides and Battery Guides.

Safety

- Never let a pack's two main contacts touch each other or anything metal. A shorted LiPo can start a fire in seconds.
- Check polarity before you plug anything in, and never force a plug.
- If a pack or connector smokes, sparks or gets hot, unplug it if you can do so safely, move away, and notify Steve (lab manager) or a TA.

Ratings

Published ratings disagree: hobby vendors usually quote the peak figure as the rating. Size for continuous draw with the lower number.

Connector	Current rating	Wire gauge (AWG)	Use and notes
XT30	15 A continuous, 30 A peak	16 to 18	Small UAVs and low-power loads
XT60	30 A continuous, 60 A peak	12 to 14	General default for mid-size packs
XT90	40 A continuous, 90 A peak	10 to 12	Large packs. The lab's 6S packs and the ISDT charger's adapter use XT90
Deans (T-plug)	15 A continuous, 30 A peak. Vendors quote up to 60 A	12 to 18	Older standard. Horizon Hobby advises against it above 3S
EC3	60 A (vendor figure)	12 to 14	Bullet contacts in a blue housing

Connector	Current rating	Wire gauge (AWG)	Use and notes
EC5	120 A (vendor figure)	10 to 12	High-current packs and large vehicles
Anderson Powerpole PP15 / PP30 / PP45	15 A / 30 A / 45 A (UL rating)	16 to 20 / 12 to 16 / 10 to 14	Genderless and stackable. One housing takes all three contact sizes
Tamiya	Up to 10 A	Not stated in the sources	Old low-power packs. Not recommended for LiPo
Molex Mini-Fit Jr.	9 A per circuit at most, 6 A on the largest housings	16 to 24	Board and accessory power, not drive power
JST-XH	3 A	22 to 30	Balance leads only. Never power a load from it

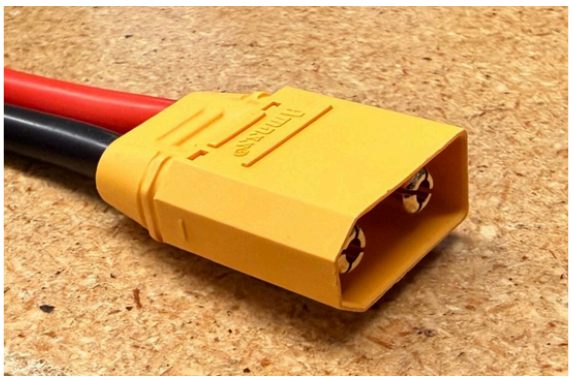
Sources: Digi-Key XT and Deans comparison for the continuous and peak figures, Ampow wire gauge chart for the vendor figures and wire gauges, Anderson Powerpole datasheet, Horizon Hobby connector guide, TTI Mini-Fit Jr. page, Digi-Key JST XH page.

Working with connectors

- The connector, the wire and the solder joint each limit the current. The weakest one sets the rating.
- Battery side is always the shrouded contact, so a loose pack cannot short on metal.
- Solder XT and Deans connectors while they are plugged into their mate. Heat alone can deform the housing and shift the contacts (Digi-Key).
- Cover XT30, XT60 and Deans solder joints with heat shrink. Only the XT90 comes with a rear cover.

In the lab

The XT90 plug is used on the lab's 6S packs. Photos of the other connector types still need to be added.



Adapter box. The lab keeps one-piece adapters between XT90, XT60, Deans and Tamiya plugs. An adapter is only as strong as its weaker side, so a Tamiya adapter limits the path to about 10 A. Use one adapter, never two in a row: each adds resistance and a place to plug something in backwards.

